INCH-POUND

MIL-C-55302/179A
26 DEC 1989
SUPERSEDING
MIL-C-55302/179
15 April 1987

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 4-ROW PIN ASSEMBLIES, 100 THROUGH 684 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.

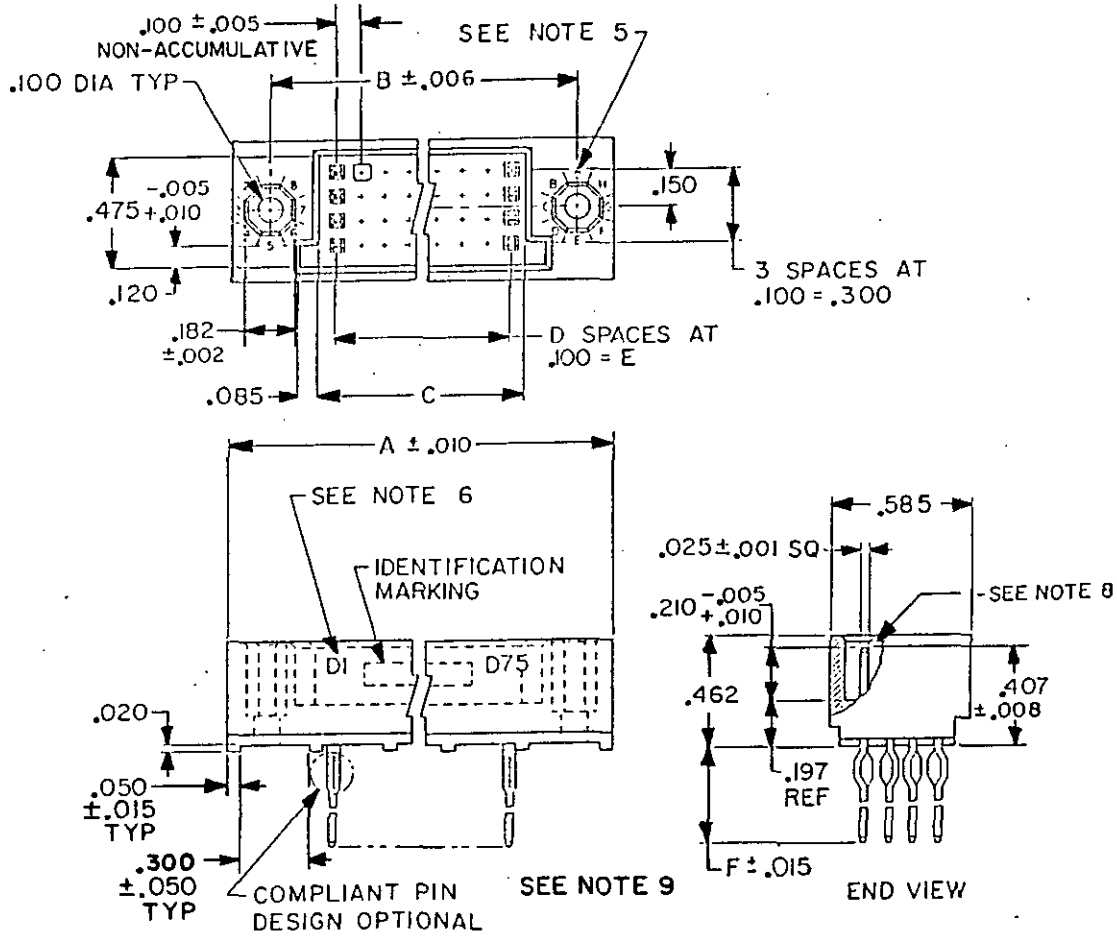


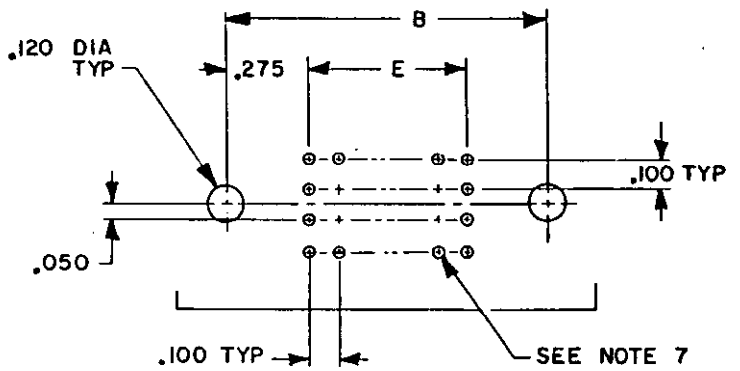
FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 100 through 300 contacts.

AMSC N/A

FSC 5935

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Recommended circuit layout

F ±.015 (0.38 mm)		Type	Type of termination
Inches	mm		
.733	18.62	I	Compliant pin (wrappost)
.533	13.54	II	
.250	6.35	IV	
.190	4.83	VI	Solder post (see note 10)

Inches	mm	Inches	mm
.001	0.03	.120	3.05
.002	0.05	.150	3.81
.005	0.13	.182	4.62
.006	0.15	.197	5.00
.008	0.20	.210	5.33
.010	0.25	.250	6.35
.015	0.38	.275	6.98
.020	0.51	.300	7.62
.025	0.64	.407	10.34
.050	1.27	.462	11.73
.085	2.16	.475	12.06
.100	2.54	.585	14.86

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/180 and /185.
5. Location indicators embossed on surface.
6. Numbers indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .042-.050 (1.07-1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ±.0010 (1.151 ±0.025 mm) drilled hole (1.15 mm drill) .001-.003 (0.03-0.08 mm) thick copper on wall, .0003 (0.008 mm) minimum tin lead plating, finished hole diameter .037-.043 (0.94-1.09 mm) after plating, .036-.043 (0.91-1.09 mm) after reflow, copper hardness (knoop) 150 maximum.
8. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ±5° (see figure 1, 2, or 3).
9. There shall be a minimum of two standoffs per inch of connector length for the balance of the part. Standoffs shall not compromise the terminating area washout feature.
10. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 100 through 300 contacts - Continued.

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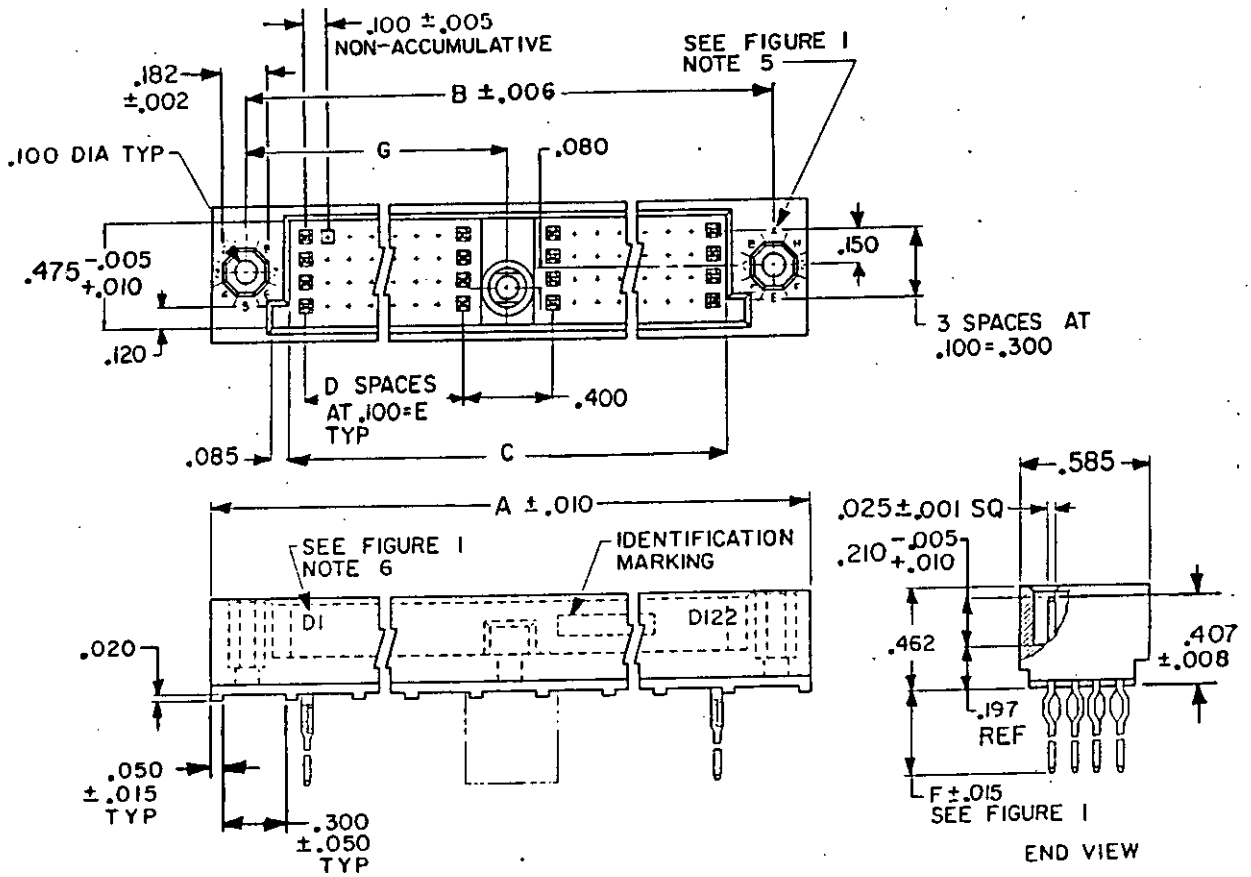
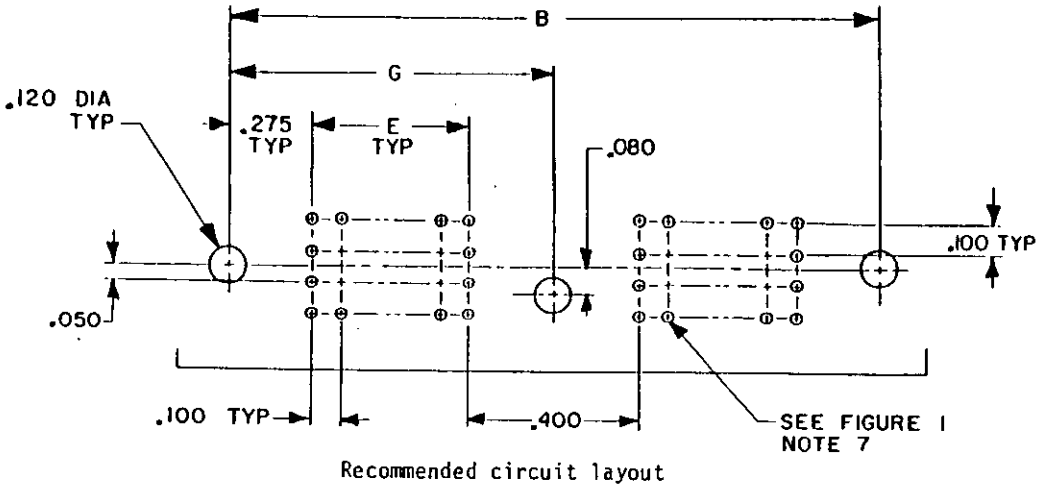


FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts.

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Recommended circuit layout

Inches	mm	Inches	mm
.001	0.03	.120	3.05
.002	0.05	.150	3.81
.005	0.13	.182	4.62
.006	0.15	.197	5.00
.008	0.20	.210	5.33
.010	0.25	.275	6.98
.015	0.38	.300	7.62
.020	0.51	.400	10.16
.025	0.64	.407	10.34
.050	1.27	.462	11.73
.080	2.03	.475	12.06
.085	2.16	.585	14.86
.100	2.54		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With one guide, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing, with one guide, 320 through 488 contacts - Continued.

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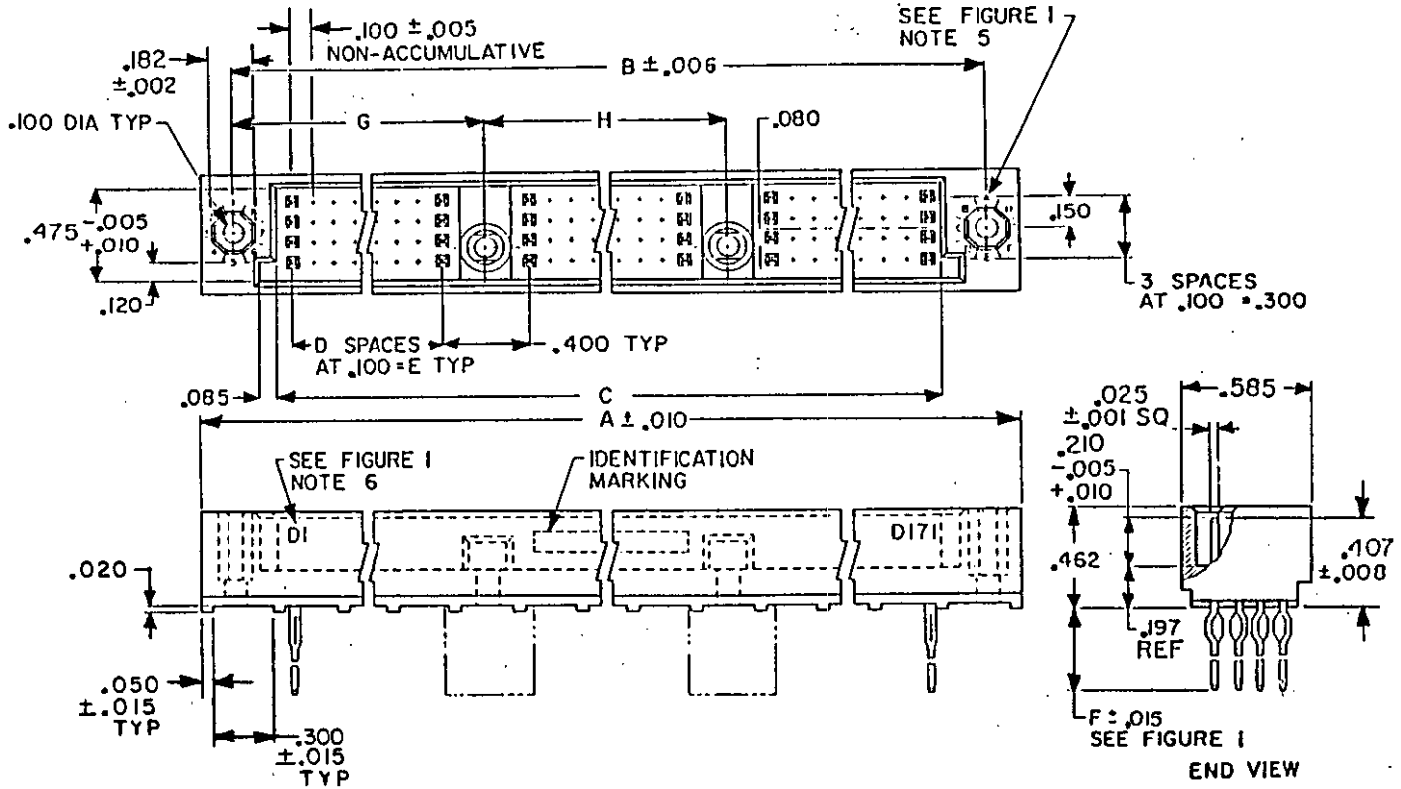
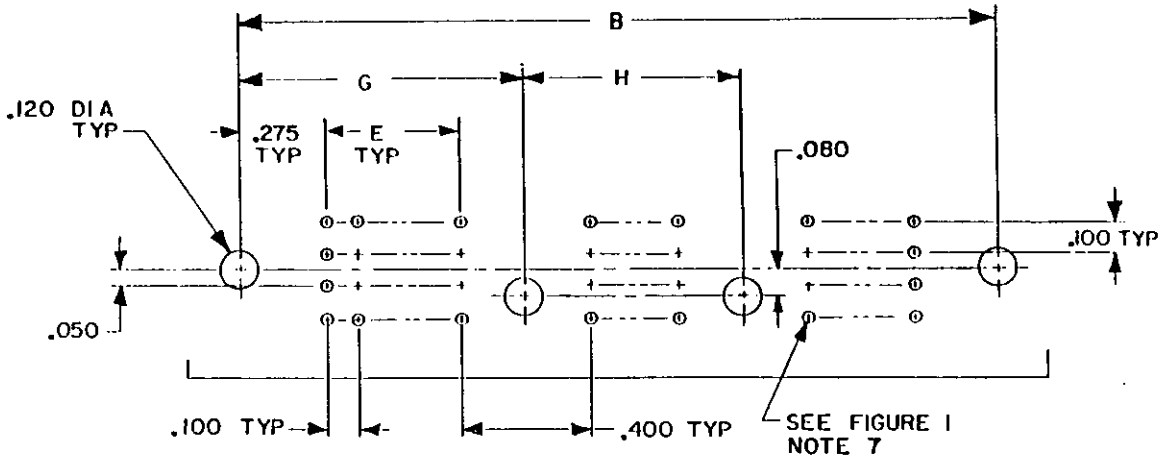


FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts.

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Inches	mm	Inches	mm
.001	0.03	.120	3.05
.002	0.05	.150	3.81
.005	0.13	.182	4.62
.006	0.15	.197	5.00
.008	0.20	.210	5.33
.010	0.25	.275	6.98
.015	0.38	.300	7.62
.020	0.51	.400	10.16
.025	0.64	.407	10.34
.050	1.27	.462	11.73
.080	2.03	.475	12.06
.085	2.16	.585	14.86
.100	2.54		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With two guides, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing, with two guides, 516 through 684 contacts - Continued.

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TABLE I. Dash number, number of contacts, contact type, and dimensions. 1/

Dash no.	No. of con-tacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 02 04 06	100	1	I II IV VI	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)			1-25
19 20 22 24	128	1	I II IV VI	3.950 (100.33)	3.650 (92.71)	3.250 (82.55)	31	3.100 (78.74)			1-32
37 38 40 42	160	1	I II IV VI	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)			1-40
55 56 58 60	200	1	I II IV VI	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)			1-50
64 65 67 69	220	1	I II IV VI	6.250 (158.75)	5.950 (151.73)	5.550 (140.97)	54	5.400 (137.16)			1-55
73 74 76 78	240	1	I II IV VI	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)			1-60
82 83 85 87	260		I II IV VI	7.250 (184.15)	6.950 (176.53)	6.550 (166.37)	64	6.400 (162.56)			1-65
91 92 94 96	268		I II IV VI	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)			1-67
100 101 103 105	280	1	I II IV VI	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)			1-70

See footnote at end of table.

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TABLE I. Dash number, number of contacts, contact type, and dimensions 1/ - Continued.

Dash no.	No. of con-tacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
109 110 112 114	300	1	I II IV VI	8.250 (209.55)	7.950 (201.93)	7.550 (191.77)	74	7.400 (187.96)			1-75
127 128 130 132	344	2	I II IV VI	9.650 (245.11)	9.350 (237.49)	8.950 (227.33)	42	4.200 (106.68)	4.675 (118.75)		1-86
154 155 157 159	416	2	I II IV VI	11.450 (290.83)	11.150 (283.21)	10.750 (273.05)	51	5.100 (129.54)	5.575 (141.61)		1-104
181 182 184 186	488	2	I II IV VI	13.250 (336.55)	12.950 (328.93)	12.550 (318.77)	60	6.000 (152.40)	6.475 (164.47)		1-122
190 191 193 195	516	3	I II IV VI	14.250 (361.95)	13.950 (354.33)	13.550 (344.17)	42	4.200 (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
199 200 202 204	540	3	I II IV VI	14.850 (377.19)	14.550 (369.57)	14.150 (359.41)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135
226 227 229 231	612	3	I II IV VI	16.650 (422.91)	16.350 (415.29)	15.950 (405.13)	50	5.000 (127.00)	5.475 (139.07)	5.400 (137.16)	1-153
235 236 238 240	636	3	I II IV VI	17.250 (438.15)	16.950 (430.53)	16.550 (420.37)	52	5.200 (132.08)	5.675 (144.15)	5.600 (142.24)	1-159
253 254 256 258	684	3	I II IV VI	18.450 (468.63)	18.150 (461.01)	17.750 (450.85)	56	5.600 (142.24)	6.075 (154.31)	6.000 (152.40)	1-171

1/ Deleted dash numbers in the series 01 through 261 are canceled with no replacement.

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TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish: Contact engagement area, .170 inch minimum length, shall be gold over nickel in accordance with MIL-C-55302. Compliant pin area (contact type I, .748 inch minimum length; type II, .548 minimum length; type IV, .265 inch minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302. Solder post area contact type VI, .205 minimum length shall be tin-lead over nickel in accordance with MIL-C-55302.

1/ Type C finish is deleted and superseded by type A or B.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, 3, and table I.

Material: In accordance with MIL-C-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, Part or Identifying Number (PIN) MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Jackscrews (see MIL-C-55302/182):

Use M55302/182-11, -12, -13, -14, -15, or -16. Jackscrews are purchased separately.

Compliant section shall meet the requirements of MIL-STD-2166, type III. For use with .093 minimum thick printed wiring boards. Sequence of tests and number of samples shall be in accordance with MIL-STD-2166, table III.

Wrappost and contacts shall be in accordance with MIL-STD-1130.

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Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .125; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

Part or Identifying Number (PIN): M55302/179 (and dash number from table I and type from table II).

Example: M55302/179-01A.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, SH
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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